



Hands-On Activity: Create a custom data table



Activity Overview

In this activity, you will use a spreadsheet to build a custom data table and analyze your data with functions. To begin, suppose you're a data analyst working for a recruiting agency. This recruiting agency helps all sorts of companies find skilled people to fill open data analytics jobs. The agency has collected data about job applications for opportunities posted on its website over the course of a year.

Scenario

Review the following scenario. Then complete the step-by-step instructions.

The agency has asked your team to optimize its online application process. Your assignment is to summarize the agency's job application data. In particular, you want to answer the following questions:

- What was the total number of applications received each month?
- How many applications were received?
- In which months were the lowest and highest number of applications received?
- What was the average number of applications received per month?

To do this, you'll work with a spreadsheet. You'll use spreadsheet functions to make calculations based on your data and create a custom data table to summarize your results.

By the time you complete this activity, you will be able to import a spreadsheet file, sort data, create a custom data table, and use spreadsheet functions to work with your data. Spreadsheets are an essential tool for every data analyst. Using spreadsheets to organize and analyze data is an important skill that you will continue to develop throughout your career.

Step-By-Step Instructions

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

Step 1: Access the spreadsheet

To use the spreadsheet for this course item, select the link below, then select the “Use Template” button to open your own version of the spreadsheet.

Note: This dataset is large, so it may take a few seconds for the rows to appear.

Link to template: [Untitled Spreadsheet](#)

OR

Download the data:

[Untitled spreadsheet](#)

[XLSX File](#)

Step 2: Understand the data

The agency's data contains information about all of the data analytics job applications received. The data includes the following column headers: **Applicant ID**, **Date**, **Job Title**, **Job Location**, **Hired**, and **Easy Apply**. Below is a description of each column header and sample values.

Column Name	Column Description	Sample Data

Applicant ID	Unique identifier for the applicants	11578773
Date	The date and time each application was received	1/1/2023 0:01:00
Job Title	The data analytics position applied for	PHARMACEUTICAL DATA ANALYST

Job Loca tion	Where the job is located	Lima, Peru
Hire d	Indicates if an applicant was hired	TRUE
Easy Appl y	TRUE if the application was submitted directly on the agency's website; FALSE if the application was downloaded and submitted via email	TRUE

Step 3: Sort your data

Because you want to answer questions based on a specific timeframe (in this case, applications received per month in 2023), start by sorting the data by date. **Sorting** involves arranging data into a meaningful order to make it easier to understand, analyze, and visualize. Considering the order in which each application was received can help you discover trends in data analytics job applications.

1. First, rename your spreadsheet. Select **Untitled Spreadsheet** and enter a new name. Use **data_analyst_jobs_2023** or a similar name that clearly describes the data your spreadsheet contains.
2. When you are working in a spreadsheet, you can have multiple sheets open. Currently, your spreadsheet contains one sheet labeled **2023_data_analyst_job**. Rename this sheet by selecting the sheet tab and choosing **Rename** from the menu. Then, enter **raw data**.

17	1159	
18	1168	Delete
19	1181	Duplicate
20	1166	Copy to
21	1183	Rename
22	1186	Change color
23	1166	Protect sheet
24	1163	Hide sheet
25	1190	View comment:
26	1186	
27	1172	
28	1183	
29	1193	
30	1182	
31	1166	Move right
32	1184	Move left
33	1158	

+ ≡ raw data ▼

3. Make the columns **Job Title** (C) and **Job Location** (D) wider by dragging the right boundary of the column headings.

4. Select all the data in the spreadsheet by selecting the cell where the rows and columns intersect.

5. From the menu bar, select **Data > Sort range > Advanced range sorting options**.

6. In the pop-up window, select the **Data has header row** box.

7. In the **Sort by** dropdown, choose the header **Date**. Then, select **A to Z** to sort in ascending order.

8. Finally, select **Sort**.

Your spreadsheet now displays job applications received in chronological order.

Step 4: Create a custom data table

Now that you've sorted your data, you're ready to create a custom data table to help you answer each of your questions. Your table will clearly summarize the data. Plus, if you want to share your results, your table will be well-organized and easy to understand.

Count the number of applications received each month

First, use spreadsheet functions to help you find the total number of applications received in each month.

1. To start, select the **Add sheet** icon (the plus sign) in the menu bar to add a new sheet to your spreadsheet. You'll create your data table in this sheet

2. Rename the new sheet. Select the sheet tab and choose **Rename** on the menu. Then, enter **summary data**.

3. Next, add column headers to your table. In cell A1 of your **summary data** sheet, enter **Month**. In cell B1, enter **Applications**.

4. Underneath the **Month** label in cell A2, enter **January**. Press **Enter**.

5. Now, use autofill to add the rest of the months of the year. Select cell A2 again. The fill handle will appear in the cell. Select on the fill handle and drag it down to cell A13 to autofill all the months of the year.

6. Next, convert the number values in the **Date** column in the **raw data** sheet into text. Select the **raw data** tab to return to the **raw data** sheet. In cell G1, enter **Month**.

7. The **TEXT** function converts a number into text according to a specified format. In this case, list them in which an application was received. Use the format “mmmm” for the full name of the month. In cell G2, enter the following code (do not copy+paste):

```
=TEXT (B2 , "mmmm")
```

The first entry **B2** refers to the cell you want to convert. The second entry **("mmmm")** refers to the specific format you want to use. Press **Enter**.

(Note: It is *very important* that you manually enter all formulas and functions. They should not be copied and pasted from the activity, as this will result in an error message.)

8. If a box pops up with the option to autofill the column, select the check mark or enter **Ctrl + Enter** (Windows) or **Cmd + Return** (Mac). If this box does not pop up, select cell G2. Then, **double-click** on the fill handle to copy the function down the column. This will populate all the cells in the column with the corresponding month.

9. Now you're ready to total the applications by month. You could do this manually, by filtering the data and counting the number of entries for each month, but this would take a long time and be prone to errors. Instead, use the **COUNTIF** function.

10. The **COUNTIF** function quickly counts how many items in a range of cells meet a given criterion. First, select the **summary data** tab to return to your **summary data** sheet. Then, in cell B2, enter **=COUNTIF('raw data'!G:G,A2)**. The first entry **'raw data'!G:G** refers to the range where you are counting the data. The range is located on your **raw data** sheet **'raw data'!** and includes all column G **G:G** entries. This column contains the data for months. The second entry **A2** refers to the criterion you want to count. In this case, it's "January," the value in cell A2 of your **summary data** sheet. The function calculates how many times **January** (the criterion) appears in the **Month** column (the range).

11. Press **Enter**. You'll notice the value 2387 appears in cell B2. This means that 2,387 job applications were submitted in January.

12. Select cell B2. **Double-click** the fill handle to copy the function down through cell B13.

Now your table lists the total number of applications submitted for each month:

Month	Applications
January	2387
February	2312
March	2536

April	2544
May	2954
June	2990
July	3138

August	2969
September	2865
October	2751
November	2508

December	2642
----------	------

Step 5: Find the total number of applications received

Now that you've calculated the number of applications received in each month, use spreadsheet formulas to calculate the total number of applications received.

1. Label the cell in which you'll calculate your result. In cell A14, enter **Total**.
2. In cell B14, enter `=SUM(B2:B13)`. This function calculates the number of applications received from January through December.
3. Cell B14 contains the total number of applications, 32596.

Step 6: Find the months with the lowest and highest number of applications received

Use the **MIN** and **MAX** functions to calculate this information.

1. First, make labels for your results. In cell A16, enter **Min**. In cell A17, enter **Max**.
2. The **MIN** function returns the minimum value in a numeric range. In cell B16, enter **=MIN (B2 : B13)**. The result, 2312, is the lowest number of applications received in any month in 2023.
3. The **MAX** function returns the maximum value in a numeric dataset. In cell B17, enter **=MAX (B2 : B13)**. The result, 3138, is the highest number of applications received in any month in 2023.

Step 7: Find the average number of applications received per month

Use the **AVERAGE** function to calculate this information.

1. First, make labels for your results. In cell A18, enter **Avg**.
2. The **AVERAGE** function returns the average value in a numeric dataset. In cell B18, enter **=AVERAGE (B2 : B13)**. The result, 2716.33, is the average number of monthly applications received in 2023.

Your work will help your team discover important trends and patterns in the agency's data and generate insights for optimizing the application process. For example, because your findings reveal that February was the slowest month, the agency can devote more of its advertising and outreach budget to February and less to the peak month of July. This is the strategic impact of data analysis.

Optional Step 8: Explore formatting options

Feel free to explore formatting options for your data table using bold, center align, fill color, borders, and more. Formatting lets you highlight important information and helps capture the attention of your audience.

Pro Tip: Save the activity template

Be sure to save a copy of the template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

1.

Question 1

Reflection

Which of the following functions counts how many items in a range of cells meet a given criterion?

Answer :

COUNTIF

TEXT

MAX

SUM

Correct

The **COUNTIF** function quickly counts how many items in a range of cells meet a given criterion. Using functions to make calculations and analyze data is an important skill for a data analyst. Going forward, you will continue to develop this skill as you work with more complex datasets.

2.

Question 2

In this activity, you learned how to analyze data using spreadsheet functions. In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- How does using functions help you quickly gain insights into large amounts of data?
- How does creating a data table help you organize and communicate important aspects of your data?

Answer:

When we apply a function that is a one-line command, it can calculate a huge amount of data, just like in this scenario, where the COUNTIF function counted 32956 numbers in just a second, and in a second we can find how many candidates gave the application for the job, so it gives a super-fast result for insight when we use a function.

A data table is a very important thing for finding insights from any data. because we can clearly and methodically figure out which and what type of data is needed for the insights. Through a data table, we can easily organize and find the true fact of our data and communicate important aspects of data for the insights. It's organized through a table so we can visualize our data easily, so it's very important.

Correct

Congratulations on completing this hands-on activity! In this activity, you imported a dataset, sorted your data, created a data table, and used functions to make calculations and analyze your data.

An effective response would note that spreadsheet functions help data professionals quickly analyze large amounts of data. It would also mention that creating data tables can help summarize and communicate important aspects of data. In upcoming activities, you will continue to explore the ways in which spreadsheet functions can help you work with complex datasets.